

Unit-1

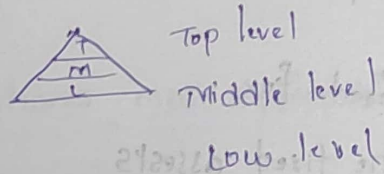
Nature & Scope of Management Economics

Management economics is the process of integrating economic theory with business practice for the purpose of facilitating decision making and forward planning by the management.

Nature of Managerial Economics

- close to micro economics
- Normative Statements
- problem solving in nature
- Alternative solutions
- inter disciplinary nature
- operates backdrop against macro economics
- Assumptions and limitations

Management: Omnipresent; present everywhere) accessing the decisions by the people.



Managerial
Economics
techniques
and concepts



Managerial decisionary

Demand analysis

Production analysis

Cost analysis

Determination of price of product

Profit of planning

Investment decisions

Make (or) Buy decisionary

→ Essentials and Limitations :
Assumptions

*

BEP (Break-Even point)

where Total Revenue = Total cost

Break-Even Analysis (low sales and all)

Demand analysis :

10/04/2023

- i) Law of demand
- * short term (< 1 year)
 - * long term (more than 1 year)

4 M's (Man, Money, Machine, Material)

Capital

- i) Fixed capital ii) Working capital

↓
Amount required for
converting the fixed assets
into amount. (Time 1 year)

- fixed asset : Time required to convert into amount is 1 year
(more than 1 year)
- current asset : Time required is 1 year.

↓
current liability (CL)

↳ Amount yet to be paid.

- If time is above 1 year long term liability
- If time is below 1 year current liability

ii) Human resource

Process of analyzing demand & Supply of the Man-power.
(Estimating)

iii) constraints for the product demand

- ① Desire on the part of the buyer
- ② Ability to pay
- ③ Willingness to pay for it

Law of demand states that when the other things stays constant, the quantity demanded of a given product (or) service rises with every fall in price and decreases with every rise in price and vice versa.

[only on some assumptions not applicable for every]

- i) price of the product must be constant
- ii) Taste of the references must be constant
- iii) No change in future income levels of the consumer.
- iv) No change in future expected price of the product.

Exceptions in law of demand:

- i) law of demand is not applicable when there is scarcity of necessities

- ii) If a paradox goods (luxury) veblen goods (luxury)
- iii) Giffen paradox goods (daily products)
- iv) Where there is an ignorance of prices by the costumers.

Demand determinance :

The word determinants refers the "factors",

- i) price
- ii) (desire) income level of consumer
- iii) references & tastes
- iv) future expected price $\rightarrow$ future expected changes in
- v) size of population $\rightarrow$ income levels of consumer
- vi) Advertising & Marketing $\rightarrow$ change in price of
- chanel distributions. $\rightarrow$ substitutes & complementaries.
- Any other factor which influences the quantity demanded of a given product (or) Service.

chand : i) Manufactur to consumer

ii) M - W - consumer
(whole salers)

iii) M - W - R - consumer
(Retail)

Demand Elasticity : $\frac{Q_2 - Q_1}{Q_1} = \frac{10 - 5}{5} = 1$

refers to the rate of response at which the quantity demanded of a given product with price and any other demand determinant.

11 Oct 2023

$$E_d = \frac{\% \text{ (or) proportionate change in Quantity demanded of product } x}{\% \text{ (or) proportionate change in price (or) any other demand determination}}$$

Types :

i) price elasticity of demand

ii) Income

iii) Cross

iv) Advertising

$$E_d = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}}$$

Q - quantity

P - price

It refers to the percentage of proportionate change in quantity demanded of product as (x) to percentage of proportionate change in price of product (x).

ii) Income elasticity of demand:

refers to the percentage (or) proportionate change in quantity demand of product (x) to percentage of proportionate change in income levels of the consumers.

$$Ed_1 = \frac{Q_2 - Q_1}{Q_1}$$

$$\frac{P_2 - P_1}{P_1}$$

iii) Cross elasticity of demand :

$$Ed_c = \frac{Q_2 - Q_1}{Q_1} \cdot \frac{P_y - P_x}{P_x}$$

Product x

Product y

iv) Advertising elasticity of demand :

refers to the percentage or proportionate change in quantity demand of product (x) to the percentage or proportionate change in the demand of Advertising budget.

$$Ed_A = \frac{Q_2 - Q_1}{Q_1} \cdot \frac{A_2 - A_1}{A_1}$$

A : advertising budget

(Perfectly)

Measurement of price elasticity of demand :

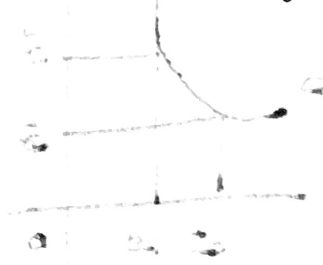
i) perfectly elastic demand :

No change or small change in price leads to indefinite change in quantity demanded of a given product or service is said to be perfectly elastic demand.

$$Ed_p = \frac{\% \Delta Q_d}{\% \Delta P}$$

ii) perfectly inelastic demand: when a significant change in price leads to no change in quantity demanded is said to be perfectly inelastic demand.

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} = 0$$



iii) Relatively elastic demand:

Demand is said to be relatively elastic when change in quantity demanded is more than the change in price.

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} > 1$$

iv) Relatively inelastic demand:

Demand is said to be relatively inelastic when the change in quantity demanded is less than the change in price.

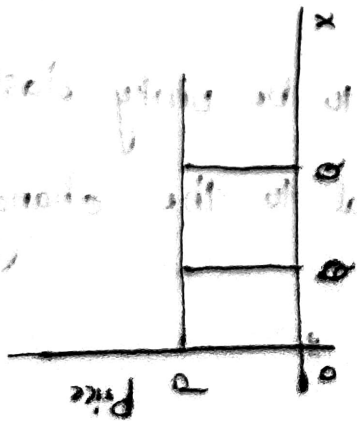
$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} < 1$$

v) unity elasticity:

Demand is said to be unity elasticity when change in quantity demanded is equal to the change in price.

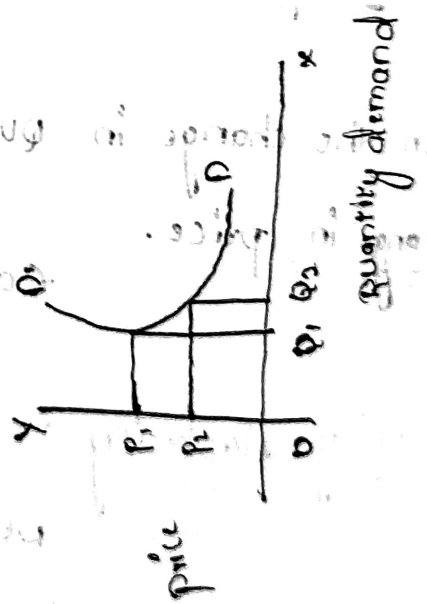
$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} = 1$$

Perfectly elastic demand



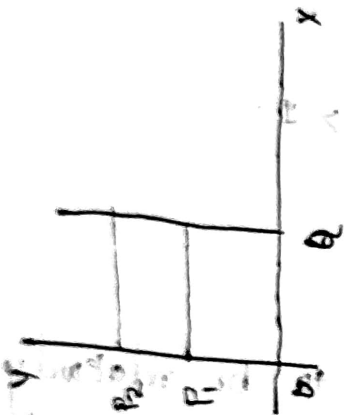
quantity demand

Relatively inelastic demand



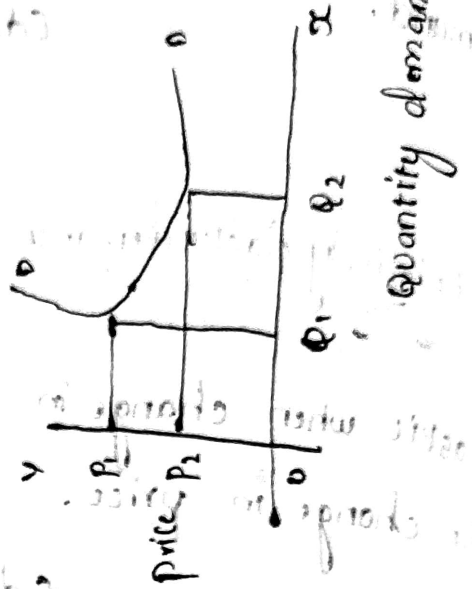
quantity demand

perfectly inelastic demand



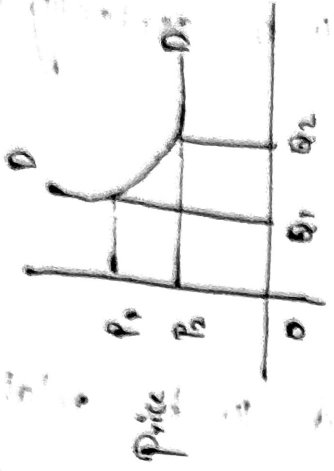
quantity demand

unity elasticity



quantity demand

Relatively elastic demand



quantity demand

Problems on elasticity of demand:

1. The Quantity demanded for product X is 1000 units at a price of ₹ 100 and the price falls to ₹ 90, the quantity demanded increased to 1500 units.

$$\text{Sol: } Ed_p = \frac{\% \Delta Q_p}{\% \Delta p} = \frac{Q_2 - Q_1}{Q_1} \times \frac{P_2 - P_1}{P_1} = \frac{1500 - 1000}{1000} \times \frac{90 - 100}{100}$$

$$= \frac{500}{1000} \times \frac{-10}{100} = \frac{1}{2} \times \frac{-10}{100} = -5$$

Here the Ed_p is less than 1.
So, it is Relatively inelastic demand.

2. Determine the advertising elasticity of demand given that the quantity demanded for product X is 1,00,000 units at a monthly advertising budget of ₹ 10,000. The monthly advertising budget slashed to 5000 and the quantity demanded will fall down 30,000 units per day.

$$\text{Sol: } Ed_p = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{-30,000}{1,00,000} \times \frac{10,000}{5000 - 10,000}$$

$$= \frac{-30}{100} \times \frac{10}{-5} = \frac{3}{1} = 3$$

It is a Relatively elastic demand.

3. Price of coffee increases from ₹ 50 per kg to ₹ 70 per kg.
As the result, Demand for tea increases from 25 kg to 30 kg.
cross elasticity of demand for tea for coffee.

Sol:

$$Ed_p = \frac{10 - 5}{5} = \frac{5}{5} = 1 \times \frac{5}{2} = 2.5$$

$$\frac{1000 - 1000}{1000}$$

$$\frac{10 - 5}{50}$$

$$\frac{25}{50}$$

It is a Relatively elastic demand.

4. Quantity demanded of Smartphones increases from 10.00 to 12.00 when consumers income increase by 10%. Calculate income elasticity of demand.

Sol:

$$Ed_p = \frac{\% \Delta Q}{\% \Delta P} = \frac{1200 - 1000}{1000} \div \frac{1}{10} = \frac{200}{1000} \times \frac{1}{10} = \frac{2}{10} = 0.2$$

It is a relatively inelastic demand.

5. If the QD for a basic food item decrease from 800 to 700 when consumers income increases by 5%. Calculate income elasticity.

$$Ed_i = \frac{700 - 800}{800} \div \frac{5}{100} = \frac{-100}{800} \times \frac{100}{5} = -\frac{25}{8}$$

P.E. = $-\frac{25}{8}$ = Relatively inelastic.

6. If the price of the Smartphone increases by 15% as the result the quantity demanded of 4 cases decreased by 8%. Evaluate the cross elasticity of demand for phone & phone cases.

sol: $\% \Delta QD_p = \frac{\% \Delta Q_{(x)}}{\% \Delta p_{(y)}} = \frac{-8}{15} = -0.53$

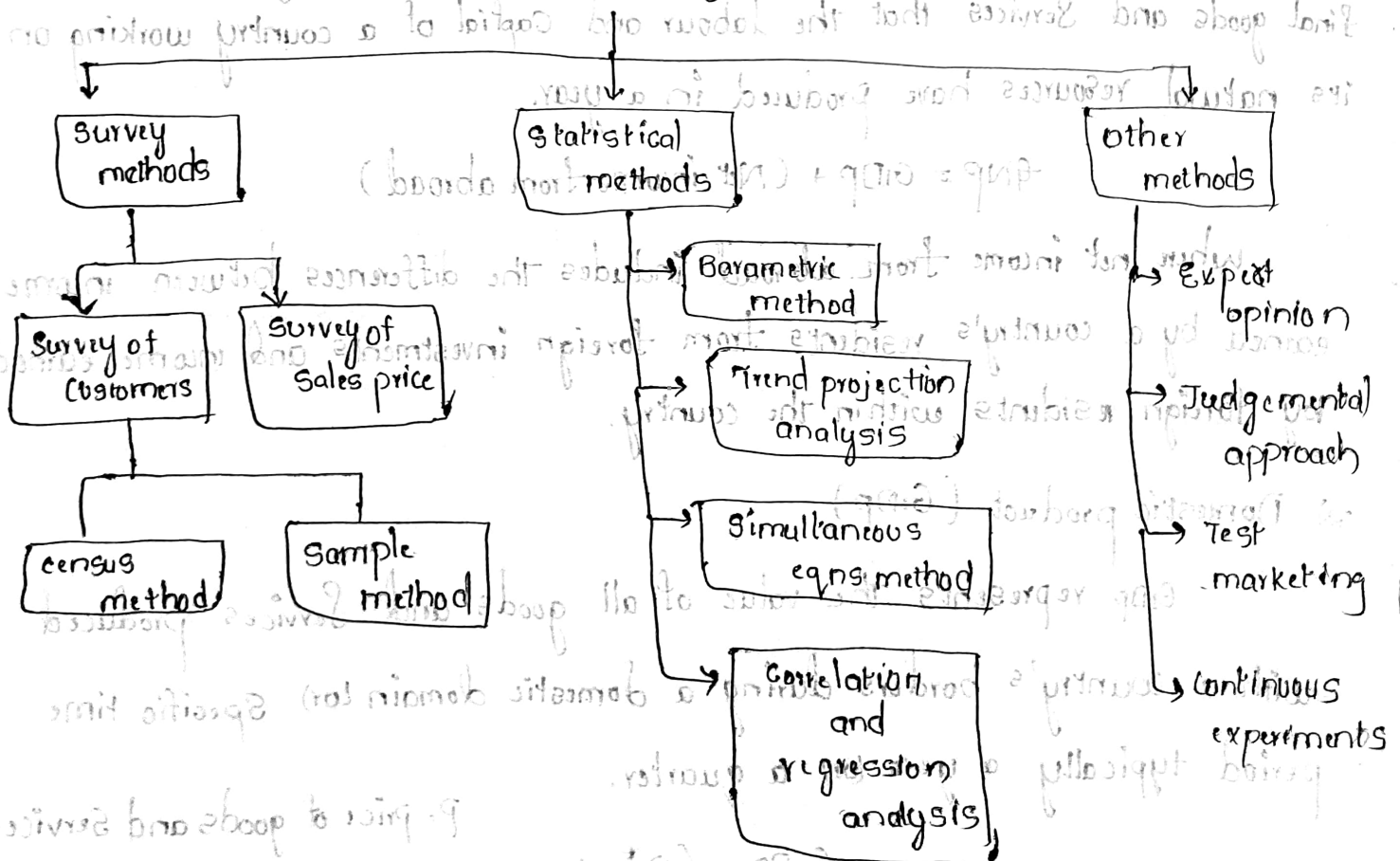
It is relatively inelastic demand

Demand forecasting

Demand forecasting is the process of predicting the future demand of the given product (or) service based on present and past data.

No company can use Census method in demand forecasting.

Demand forecasting methods



(Assignment)

National income

10 components of national income.

What is national income?

National income is the sum of all income made in the country's economy on an aggregate level. It is essential measure of economic performance. There are three methods for calculating the income of any economy.

The income approach

- The expenditure approach

- The value added approach

1) Gross National product (GNP)

GNP has been defined as the total market value of all time final goods and services produced in a year. It is the money value of all the final goods and services that the labour and capital of a country working on its natural resources have produced in a year.

$$GNP = GDP + (\text{Net income from abroad})$$

where net income from abroad includes the differences between income earned by a country's residents from foreign investments and income earned by foreign residents within the country.

2) Domestic product (GDP)

GDP represents the value of all goods and services produced within a country's borders during a domestic domain (or) specific time period typically a year (or) a quarter.

$$GDP = (P * Q)$$

P = price of goods and services
Q = Quantity of goods & services



$$GDP = C + I + G + (X - M)$$

$X - M \rightarrow$ Export minus import

C - consumption

G - government expenditure

I - investment

3) personal income

It is the total income earned by individuals within a country. It includes wages, salaries, rental, income, interest dividends, and transfer payments like Social Security and welfare.

Formula: $NNP - \text{taxes on production and imports} - \text{Social Security contributions} - \text{corporate profit taxes} - \text{undistributed corporate profits} + \text{transfer payments to individuals}$.

Net National product (NNP) at MP: It is the market value of net output of final goods and services produced by an economy during a year and net factor income from abroad.

$NNP = GDP - \text{Description}$

$NNP = C + I + G + (X - M) + NFIA - IT$ Description

$NFIA = \text{net factor income from abroad}$

$IT = \text{indirect factor}$.

4) Disposable income

It is the income left for individuals or households after taxes have been deducted. It's the money people have available for consumption and savings.

Formula: $\text{Disposable income} = \text{Personal income} + \text{personal taxes}$

5) Corporate profits:

This component includes the profits earned by business and corporations operating within the country. It may be divided into before tax and after tax profits.

Formula:

$$\text{Corporate Profits} = \text{Gross Corporate Profits} - \text{Corporate Profit Taxes}$$

6) Capital Depreciation:

This represents the amount of capital that has been worn out or used up during the production of goods and services, it is the subtracted from GDP to arrive at Net National Product (NNP).

Formula:

$$\text{Capital Depreciation} = \text{Gross Investment} - \text{Net Investment}$$

7) Indirect Business Taxes:

Taxes imposed on business such as sales taxes, excise taxes and tariffs. They are subtracted from GDP to determine the net national product.

Formula:

$$\text{Indirect Business Taxes} = \text{Business Taxes} + \text{Subsidies}$$

8) Net Exports:

Net exports refer to the difference between a country's exports (goods and services sold to other countries) and imports. A positive value indicates a trade surplus, while a negative value indicates a trade deficit.

Formula: $\text{Net Exports [Trade Balance]} = \text{Exports} - \text{Imports}$

9. Transfer payments :

Transfer payments made by the government to individuals or households for which no goods or services are exchanged. These include social welfare payments, unemployment benefits and subsidies.

Formula:

$$\text{Transfer Payments} = \text{Payment to individuals} + \text{Payment to government}$$

10. Gross and net national income

Gross national income (GNI) is similar to GDP but includes net income from abroad. NNI is GNI minus capital depreciation.

NNI reflects the net income generated within a country's borders for after accounting for wear and tear on capital.

Formula: $NNP = GNP - \text{Depreciation}$

=> measurement of national (issuc) income

=> $PCI = \text{Total national income} / \text{Total national population}$

where $PCI = \text{per capital income}$.

18/10/23

Unit-2

Concepts of production

↓
Tp

Total production

It refers to the sumtotal of all the output that can be produced at a given period of time

↓
Ap

Average production

It refers to the output that can be produced per unit of labour

↓
Mp

Marshall production

It refers to the change in Tp as a result of change in QL

$$Tp = Ap \times QL$$

$$Ap = Tp / QL$$

$$Mp = \frac{\Delta Tp}{\Delta QL}$$

QL: Quality labour

$$Mp = Tp_n - Tp_{n-1}$$

Law of returns to a factor

i) Law of increasing return to a factor

According to this law, if we increase units of variable factor which is labour in the short run, then the Tp will increase at increasing rate.

Assumptions

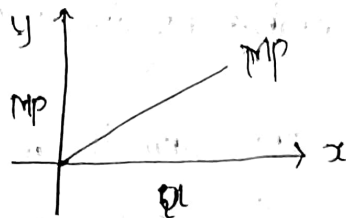
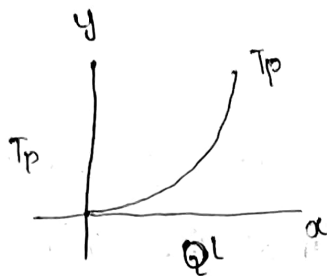
- i) Applications on short run
- ii) no change in technology
- iii) labour is of homogeneous

This law is applicable for industrial sector.

Explanation of law

QL	Tp	Mp
1	10	10
2	30	20
3	60	30
4	100	40
5	150	50



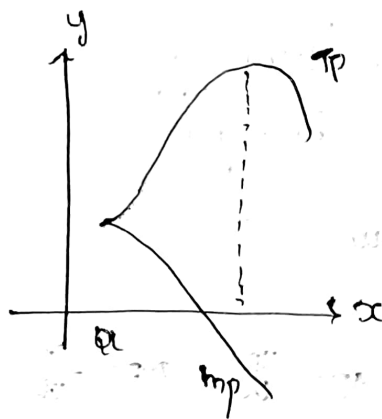


2) Law of diminishing return to a factor

According to this law, if we increase units of variable factor which is labour in the short run then T_p will increase at diminishing rate. This law is applicable for industrial sector.

Explanation of law

Q_l	T_p	M_p
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

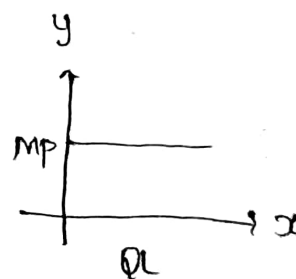
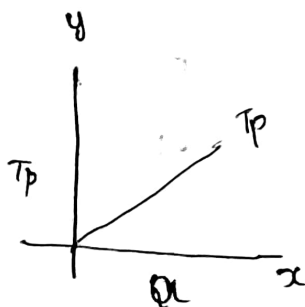


3) Law of constant return to a factor

According to this law, if we increase units of variable factor which is labour in the short run, T_p increase at constant rate.

Explanation of law

Q_l	T_p	M_p
1	10	10
2	20	10
3	30	10
4	40	10
5	50	10



*** Law of Variable Proportions (LVP)

28/10/23

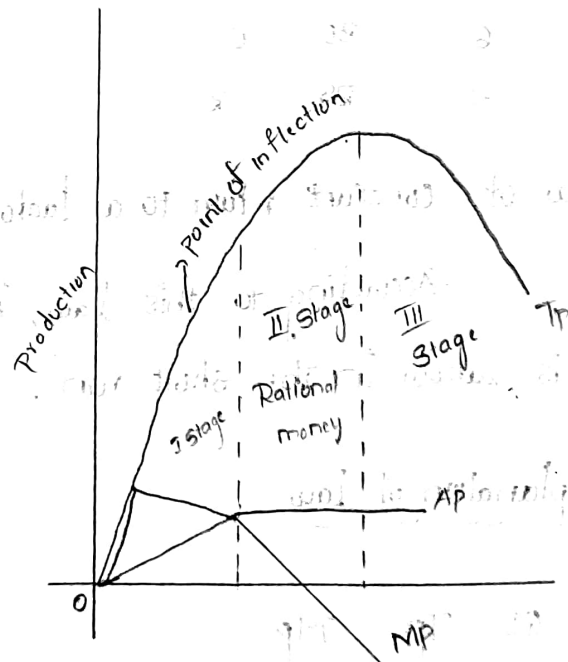
According to this law if we increase the units of variable factor which is labour in the short run. Then first of all the T_p will increase at increasing rate. After some time if u increase the variable factor which is labour, then the T_p will increase at diminishing rate.

Assumptions:

- i) Applicable for Short run.
- ii) Labour is only the variable factor
- iii) No change in technology
- iv) Labour is of homogeneous

Explanation of law:

	QL	T_p	$AP = \frac{T_p}{Q_L}$	$MP = \frac{\Delta T_p}{\Delta Q_L}$
IR	1	2	2	2
	2	5	2.5	0.5
	3	9	3	4
DR	4	12	3	3
	5	14	2.8	2
	6	15	2.5	1
	7	15	2.1	0
	8	14	1.7	-1



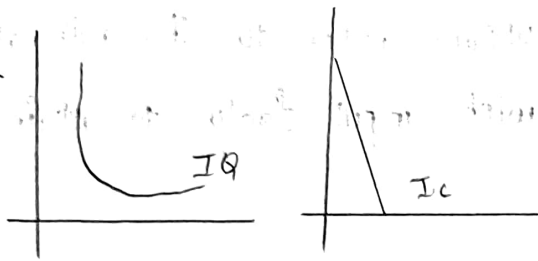
Isoquants

Iso costs

Isoquant:

An isoquant is a product indifference curve which is used to show the various combinations of input factors which is the same level of output.

(holds)



properties:

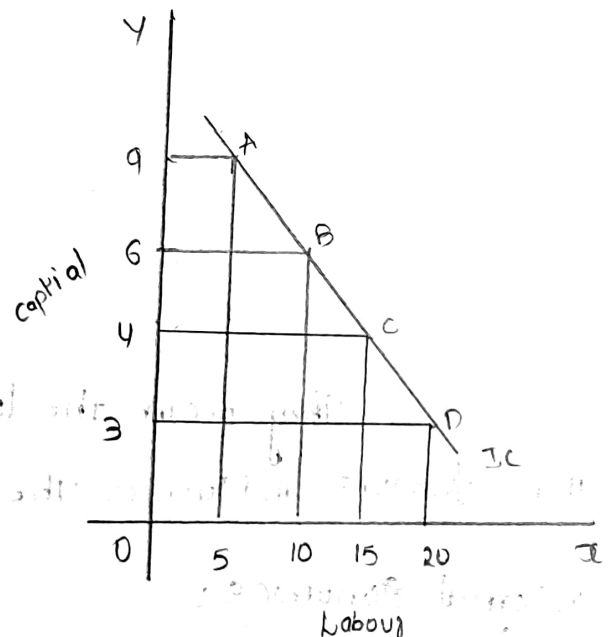
- Downward sloping
- Don't touch the axis
- Don't intersect

* Iso costs:

An isocost is a cost of which shows various combination of inputs to attain the given level of output, which costs the same amount of money for the producer.

(which costs same money & same quantity but takes different inputs (or) products)

Combination of inputs	Labour	Capital	Output (100m of cloth)
A	5	9	100
B	10	6	100
C	15	4	100
D	20	3	100



$$A = 5L + 9C = 100$$

$$B = 10L + 6C = 100$$

$$C = 15L + 4C = 100$$

$$D = 20L + 3C = 100$$

1 Nov 2023

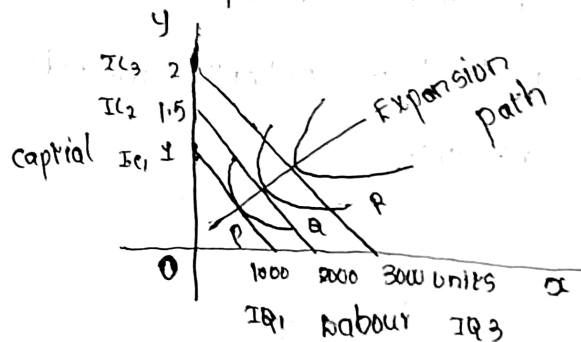
Marginal rate of technical substitutions

Marginal rate of technical substitutions refers to the rate at which how an input factor substituted with input factor to attain given level of output.

$$MRTS = \frac{\Delta K}{\Delta L}$$

→ Every producers ultimate goal is to reduce the cost of production

→ Isocosts & Isoquants are used to select various combination of input factors which costs the producer in the least cost.



Internal & External Economical Scale :

{ factors influencing the long term production

↓ which are variable in nature

↳ factors

↓ for short term

• which leads to form Scale (int & ext scale)

They occur the long term production process. These are the factors influences the production Scale in long term.

Internal Economics :

i) financial economies

ii) managerial economies

iii) Commercial economics

iv) risk economics

v) Economics of RND

External Economics:

i) Economics of welfare

ii) " " concentration

iii) " " RND (research & development)

End of product Analysis

Cost Analysis :-

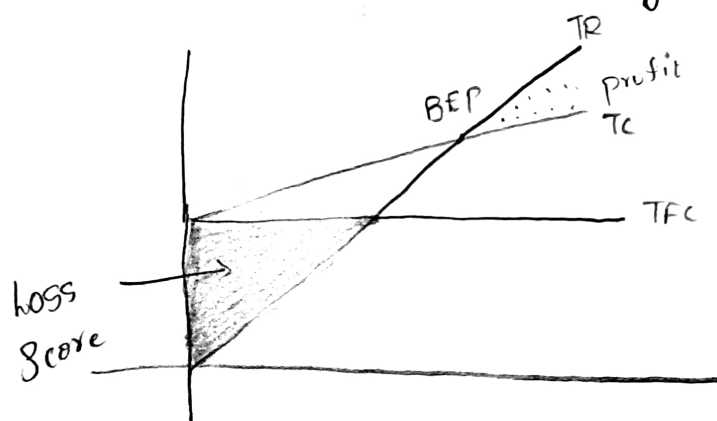
• Break Even analysis: cost volume profit analysis at different levels of sales is called Break Even Analysis.

→ Break Even point is a point at which, Total Revenue is equal to the Total cost.

$$\text{BEP: } Tc = TR$$

Assumptions:

- costs are segregated into Total fixed cost & Total variable cost
- TFC remains constant (or) unchanged.
- Selling price per unit remain constant at all levels of sales. only the volume of production changes the cost.



$$\text{BEP} \\ Tc = TR$$

Determination of BEP

- ① $BEP = \frac{F}{P/v \text{ ratio}}$ (in value) F : fixed cost
- ② $BEP = \frac{F}{C_{pu}}$ (in units)
- C_{pu} : selling price per unit - variable cost per unit
- ③ $P/v \text{ ratio} = \frac{\text{contribution}}{\text{sales}} \times 100$
- (or)
- $P/v \text{ ratio} = \frac{\text{change in profit}}{\text{change in sales}} \times 100$
- ④ $MOS = \text{Total sales} - \text{Break even sales}$
- MOS = Margin of safety
- ⑤ Sales volume to earn a desired profit = $\frac{F + D.p}{P/v \text{ ratio}}$
- ⑥ Contribution = Selling price - variable cost
- (or)
- $F + \text{profit}$

1) Determine ~~Top~~ ^{BEP} from the following information.

Year	Sales (Rs)	profit (Rs)
2020	1,50,000	20,000
2021	1,70,000	25,000

Sol:

$$P/v \text{ ratio} = \frac{\text{change in profit}}{\text{change in sales}} \times 100$$

$$= \frac{25000 - 20000}{1,70,000 - 1,50,000} \times 100$$

$$= \frac{5000}{20,000} \times 100 = \frac{100}{4} = 25\%$$

Let us consider 2021 date

$$\text{Sales} = 1,70,000 \times 25\%$$

$$= 42,500$$

contribution = Sales $\times \frac{100}{4}$

$$\text{contribution} = Sp - vc \text{ (or)} F + \text{profit}$$

$$42,500 = F + 25,000$$

$$BEP = \frac{F}{P/v \text{ ratio}}$$

$$F = 42,500 - 25,000$$

$$= 17,500$$

$$= \frac{17,500}{25\%} = 70,000/-$$

2) Given Sales is equal to ₹ 4000, variable cost is ₹ 2000, fixed cost is ₹ 1600, calculate P/v ratio, BEP volume, margin of safety profit, how many sales required to earn a desired profit at ₹ 1200

Sol:

$$\text{contribution} = \text{Selling price} - \text{Variable cost}$$

$$\text{Selling price} = 4000$$



$$\text{Contribution} = \text{Sp} - \text{VC}$$

$$= 4000 - 2000$$

$$= 2000$$

$$1) \rightarrow \text{P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$= \frac{2000}{4000} \times 100 = 50\%$$

$$2) \text{ BEP} = \frac{F}{\text{P/V ratio}}$$

$$= \frac{1600}{50\%} = 3200$$

$$3) \text{ Margin of safety} = \text{Total sales} - \text{Break even sales}$$

$$= 4000 - 3200$$

$$= 800/-$$

$$4) \text{ profit} = \text{Contribution} - F$$

$$= 2000 - 1600$$

$$= 400/-$$

$$5) \text{ Sales volume to earn a profit} = \frac{F + P}{\text{P/V}}$$

$$= \frac{1600 + 400}{50\%}$$

$$= \frac{2000 \times 100}{50} = \frac{1600 + 1200}{50\%}$$

$$= \frac{2800}{50\%}$$

$$= 5600$$

3) Determine m.o.s

Year	Sales	profit
2014	2,00,000	20,000
2015	3,00,000	75,000

Sol: $P/V \text{ ratio} = \frac{\text{change in profit}}{\text{change in sales}} \times 100$

$$= \frac{55,000}{3,00,000} \times 100$$

$$= 18.3\%$$

in year 2015, contribution = F + profit.

$$\text{Contribution} = \text{Sales} \times P/V$$

$$= 5,00,000 \times 18.3\%$$

$$= 91,500$$

$$F = \text{contribution} - \text{profit}$$

$$= 91,500 - 75,000$$

$$= 16,500$$

$$m.o.s = \text{Total sales} - BEP$$

$$= 20,00,000 - 90163.9$$

$$= 1,81,986.2$$

for 2015,

$$= 500,000 - 90163.9$$

$$= 409836.1$$

$$\text{mos ratio} = \frac{\text{mos 1}}{\text{mos 2}} = 0.4439$$

$$BEP = \frac{F}{P/V} = \frac{16,500}{18.3\%} = 90163.9$$

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- The fixed cost of a product is 20,000. Selling price per unit is ₹12 and the variable price is ₹5. Determine BEP

Sol: Given

$$\text{Fixed cost} = 20,000 ; \text{Sp per unit} = 12$$

$$C_{pu} = 12 - 5$$

$$= 7$$

$$\text{Variable cost per unit} = 5$$

$$BEP = \frac{F}{C_{pu}} = \frac{20,000}{7}$$

$$= 2857.14$$



The Sales Startover and the profit unit 2 years are as follows

Year	Sales (Rs.)	Profit
2014	1,40,000	15,000
2015	1,60,000	20,000

- a) p/v ratio b) Sales required to earn a desired profit of Rs 40,000
c) profit when Sales are 1,20,000.

Sol: Given a) $p/v \text{ ratio} = \frac{\text{Change in profits}}{\text{change in sales}} \times 100$

$$= \frac{20,000 - 15,000}{1,60,000 - 1,40,000} \times 100$$

$$= \frac{5000}{20000} \times 100 = 25\%$$

- b) Sales required to earn a desired profit of ₹ 40,000

$$\Rightarrow \frac{F + DP}{p/v \text{ ratio}}$$

$$\boxed{\text{Contribution} = p/v \text{ ratio} \times \text{Sales}}$$

(year 2015) $= 25\% \times 1,60,000$

$$= 40,000$$

$$\text{contribution} = F + \text{profit}$$

$$40,000 = F + 20,000$$

$$\boxed{F = 20,000}$$

$$\Rightarrow \frac{F + DP}{p/v \text{ ratio}} = \frac{20,000 + 40,000}{25\%} = 2,40,000$$

- c) profit when sales are 1,20,000

$$1,20,000 = \frac{F + DP}{p/v} \Rightarrow 1,20,000 = \frac{20,000 + DP}{25\%}$$

$$\boxed{DP = 10,000}$$

• SV limited furnishes the following data relating to 2 years

Particulars	Year 2019	Year 2020
Total Sales	45,000	50,000
Total costs	40,000	43,000

- a) p/v ratio b) fixed expenses c) break even points
d) Mos for 2 years

Sol:-

$$\begin{aligned}\text{profit in 2019 is} &= \text{Total Sales} - \text{Total costs} \\ &= 45,000 - 40,000 \\ &= 5,000\end{aligned}$$

$$\begin{aligned}\text{profit in 2020} &= 50,000 - 43,000 \\ &= 7,000\end{aligned}$$

$$\begin{aligned}\text{p/v ratio} &= \frac{\text{change in profits}}{\text{change in Sales}} \times 100 \\ &= \frac{2000}{5000} \times 100 = 40\%\end{aligned}$$

b) fixed expenses

$$\begin{aligned}\text{contribution} &= \text{Sales} \times \text{p/v ratio} \\ &= 45,000 \times 40\% \\ &= 18,000\end{aligned}$$

$$\begin{aligned}\text{Contribution} &= F + \text{profit} \quad (\text{profit} = 5000) \\ F &= 18,000 - 5000\end{aligned}$$

$$\text{fixed expenses} = 13,000$$

$$\text{c) Break Even point} = \frac{F}{\text{p/v ratio}} = \frac{13,000}{40\%} \Rightarrow \text{BEP} = 32500$$

d) Mos for 2019,
margin of safety

$$\begin{aligned} \text{Total Sales} - \text{Break even Sales} \\ = 45000 - 32500 \\ = 12500 \end{aligned}$$

Mos for 2020

$$\begin{aligned} \Rightarrow \text{Total Sales} - \text{BES} \\ = 50,000 - 32500 \\ = 17500 \end{aligned}$$

Year	Sales	profit
2015	1,20,000	4,500
2016	1,30,000	12,500

compute a) plv ratio b) fixed cost c) MOS d) BEP

e) variable cost for 2 years

Sol: a) plv ratio = $\frac{\text{change in profit}}{\text{change in sales}} \times 100$

$$= \frac{8000}{10,000} \times 100 = 80\%$$

b) fixed cost

$$\text{Contribution} = \text{Sales} \times \text{Plv ratio}$$

$$= 1,20,000 \times 80\%$$

$$= 96,000$$

2015

$$\text{Contribution} = \text{F} + \text{profit}$$

$$F = 96,000 - 4,500$$

$$F = 91,500$$

d) BEP = $\frac{F}{\text{Plv ratio}} = \frac{91,500}{80\%} = 1,14,375$

c) Mos for 2015 Total Sales - BEP

$$1,20,000 - 1,14,375 = 5625$$



$$\text{Ans for 2016} = \text{Total Sales} - \text{BEP} \\ = 1,30,000 - 14375 \\ = 15625$$

e) variable cost contribution = $SP - VC$

$$VC = SP - \text{contribution}$$

$$= 120000 - 96000$$

$$= 24000$$

2016
contribution = $\text{Sales} \times \text{P/V ratio}$

$$= 1,30,000 \times 80\%$$

$$= 1,04,000$$

$$VC = SP - \text{contribution}$$

$$= 1,30,000 - 1,04,000$$

$$= 26,000$$

A manufacture has sales of - ₹ 15,000 and profit of ₹ 400, if he sells 19,000 he make a profit of ₹ 1200/- find out contribution margin ratio, BEP, sales for 4400/-

Sol: Given $SP = ₹ 15,000$ profit = 400

$$SP = ₹ 19,000 \quad \text{profit} = 1200$$

$$\text{contribution margin ratio (P/V)} = \frac{\text{change in profits}}{\text{change in Sales}} \times 100$$

$$= \frac{1200 - 400}{19,000 - 15,000} \times 100$$

$$= 20\%$$

$$\text{BEP} = \frac{F}{\text{P/V ratio}}$$

$$\text{Contribution} = SP \times \text{P/V ratio}$$

$$= 15,000 \times 20\%$$

$$= 3000$$

$$\text{contribution} = F + \text{profit}$$

$$F = 3000 - 400$$

$$F = 2600$$

$$\text{BEP} = \frac{F}{\text{P/V ratio}} = \frac{2600}{20\%}$$

$$= 13,000$$

$$\text{Sales for 4400} = \frac{F + P}{\text{P/V ratio}}$$

$$= \frac{2600 + 4400}{20\%}$$

$$= 35,000$$

• A firm has fixed cost of Rs. 10,000/- & Selling price per unit is Rs. 5/- variable cost per unit is Rs 3/-

a) determine BEP in terms of unit and value

b) nos of actual sales are Rs. 9000/- units

Sol: $BEP = \frac{F}{C_{pu}}$

C_{pu} = Selling price per unit - variable cost per unit

$\Rightarrow 5 - 3 = 2$

$BEP = \frac{10,000}{2} = 5000$ - in terms of units

$P/V = \frac{C_{pu}}{SP} \times 100 = \frac{2}{5} \times 100 = 40\%$

$BEP = \frac{F}{P/V} = \frac{10,000}{40\%} = 25,000$

b) Nos = Total sales - BEP (in units)

$= 9000 - 5000$

$= 4000$

Capital



• Working capital

• Fixed capital

Assets



{ in a period of 1 year }

• Current asset

• fixed asset

↳ process of allocating financial (issue) matter in a proper manner.

$$\textcircled{1} \text{ payback period} = \frac{\text{cost of the project}}{\text{Annual cash inflows}}$$

→ B. methods

$$\textcircled{2} \text{ ARR} = \frac{\text{Average annual profit after taxes}}{\text{Average investment}} \quad \{ \text{Budget} \}$$

↳ (Investment decisions)

↓

ARR = Accounting rate of return

↓
depends upon

→ cash inflow

Average investment = $\frac{1}{2}$ of original investment (cash flow profit after taxes + Deposits)

$$\textcircled{3} \text{ NPV} = \text{PVCIF} - \text{PVCOF}$$

→ Cash outflow

(investment)

← Amount to be

NPV : Net present value method

PVCIF : present value of cash inflow

PVCOF : present value of cash outflow

NPV > 1 ✓ project accept
NPV < 1 ✗ project declined

$$\textcircled{4} \text{ profitability index} = \text{PVCIF} / \text{PVCOF}$$

$$\textcircled{5} \text{ IRR} = \text{Lower rate} + \frac{+ \text{NPV}}{\Delta \text{PVCIF}} \times \Delta R$$

IRR = Internal rate of return

↓

1) The cost of the project is 50,000/- , the cash inflows for the next 4 years are 25,000/- . what is the pay back period.

Sol:

$$\text{payback period} = \frac{50,000}{25,000} = 2 \text{ years}$$

2) ABC limited is considering 2 projects, each requires an investment of ₹ 10000. The net cash inflows from the investment in the 2 projects, X and Y are as follows.

Year	cash-inflows project X	cash-inflows project Y
1	5000	1000
2	4000	2000
3	3000	3000
4	2000	4000
5	—	5000
6	—	6000

The company has fixed 3 years pay back period as the cut off point. Advise the management which project to be accepted and why (use pay back period)

Sol:	Year	project-X	Comm. cash inflow	Year	project Y	Comm. cash inflow
	1	5000	5000	1	1000	1000
	2	4000	9000 ✓	2	2000	3000
	3	3000	12000	3	3000	6000
	4	2000	13000	4	4000	10000 ✓
	5	—		5	5000	15000
	6	—		6	6000	21000

$$\text{Pay back period} = 2 \text{ yr} + \frac{10000 - 9000}{3000}$$

$$= 2 \text{ yr} + \frac{1}{3}$$

$$= 2 \text{ yr} 4 \text{ months}$$

$$\text{Pay back period} = 4 \text{ years}$$

1st project is accepted → because of less time.

3) Firm has following information, has a project cost ₹ 81,000 and expected to generate net profits as 20,000, 25,000, 15,000, 10,000 per 4 years. calculate accounting rate of return.

Sol:

$$ARR = \frac{\text{Average annual profit after taxes}}{\text{Average investment}}$$

$$\text{Avg annual profit} = \frac{20,000 + 25,000 + 15,000 + 10,000}{4} = 17,500$$

$$\text{Avg investment} = \frac{81,000}{2} = 40,500$$

$$ARR = \frac{17,500}{40,500} \times 100$$

$$= 43.2\%$$

4) ABC limited is considering a project requires an investment of ₹ 1,00,000, the net cashinflows are as follows

Years	Cashinflows	cumm. cash	pv factor @12%	pvcif
1	30000	30,000	0.893	26790
2	35,000	65,000	0.797	27895
3	35,000	1,00,000	0.712	24920
4	40,000	1,40,000	0.636	25440
5	20,000	1,60,000	0.567	11340
Use NPV, profitability factor.				1,16,385
				1,16,385

Sol:

$$NPV = pvcif - pvcof$$

$$= 1,16,385 - 1,00,000$$

$$= 16,385$$



$$\text{profitability index} = \text{pvCIF} / \text{pvCOF}$$

$$= 116385 / 1001000$$

$$= 1.16$$

5) calculate the pay back period of the following projects each requiring cash outflow of ₹ 1,00,000. Suggest which project is acceptable

years	cash inflow of project A	comm	Cash inflow of proj B	
1	80,000	30000	30,000	30,000
2	80,000	60000	40,000	70,000
3	80,000	90,000	20,000	90,000
4	80,000	1,20,000	10,000	1,00,000
5	80,000	1,50,000	5000	1,05,000

$$\downarrow$$

$$= 3 + \frac{1,00,000 - 90,000}{30,000}$$

↓

4 Years

$$= 3 + \frac{10,000}{30,000}$$

= 3 years 4 months

project A is acceptable.

6) A company has at least two proposals for consideration. The cost of the proposals in both the cases is RS 5,00,000 each. A discount factor 12% may be used cash inflows after taxes are as under

Year	Cash inflows		pv factor @ 12%	PvCIF	
	x	y		x	y
1	1,50,000	50,000	0.893	133950	44650
2	2,00,000	1,50,000	0.797	159400	119550
3	2,50,000	2,00,000	0.712	178000	142400
4	1,50,000	1,00,000	0.636	95400	190800
5	1,00,000	2,00,000	0.567	56700	113400
				<u>Σ = 623450</u>	<u>Σ = 6,10,800</u>

$$NPV = PVCIF - pvcof$$

$$= 623450 - 5,00,000$$

$$= 1,23,450$$

↓

$$\text{Probability index} = \frac{623450}{500000}$$

$$= 1.2$$

project 'x' is preferred.

$$NPV = PVCIF - pvcof$$

$$= 6,10,800 - 5,00,000$$

$$= 1,10,800$$

$$= \frac{6,10,800}{5,00,000}$$

$$= 1.2$$

4) Determine NPV for

use cost of capital as 10 %

Year	proj(x)	proj(y)	pv	pvcif(x)	pvcif(y)
0	2,00,000	2,50,000	1		
1	50,000	40,000	0.909		
2	60,000	60,000	0.826		
3	30,000	80,000	0.759		
4	40,000	70,000	0.68		
5	20,000	75,000	0.620		
Scrap	20,000	50,000			

classification of Accounts:

Accounts are classified into 3 categories

i) personal accounts

ii) Real accounts

iii) Nominal accounts

Personal accounts:

They are of 3 types

i) Natural personal acc:

These are the accounts related to natural person.

ii) Artificial personal acc: These are the accounts related to artificial body of the person.

Ex: firm acc, bank acc, club account etc.,

iii) Representative personal acc:

These are outstanding salaries, prepaid expenses, prepaid insurance etc.,

2) Real accounts: They are of 2 types

i) Tangible real acc:

These are the accounts which can be touched, measured, purchased, sold are called as Tangible real accounts.

Ex: machinery acc, furniture acc, Sales acc, cash acc, etc.,

ii) Intangible real acc: These are the accounts which can't be touched but which can be measured in terms of money.

Ex: Patents, Good will, Trademark, Copyright etc.,

3) Nominal accounts:

Expenses, losses, incomes & gains are called as nominal accounts.

Ex: Salaries acc, wages acc, dividend, transport, electricity, discount, profit & losses etc.,

Journal

It is the primary book of accounts in which all the transactions is to be recorded in journal first, in a chronological order.

Rules of Journal:

* Personal accounts → Dr. the receiver

Cr. the Giver

* Real accounts → Dr. what comes into the business
Cr. what goes out from the business

* Nominal accounts → Dr. Expenses and losses

Cr. Incomes and gains

1) Journalize the following Transactions in the book of Adithyaram per month of march 2023.

1-march, Aditya comes into business with a cash of ₹ 5 lakhs.

- 2nd march, deposited in bank 2 lakhs.
- 5th march, purchase goods for cash 50,000/-
- 10th march, sold goods for cash 1,00,000/-
- 20th march, salaries paid 50,000/-
- 31st march, rent paid to the landlord 20,000/-

Journal book of Aditya for the month of march 2023

Date	Particular	LF	Dr	Cr
2023 mar-1	cash a/c Dr To aditya capital a/c (being business commenced with cash)		5,00,000	50,00,000
mar-02	Bank a/c --- Dr To cash a/c (being cash deposited on bank)		2,00,000	2,00,000
mar-05	purchase a/c --- Dr To cash a/c (being goods purchased for cash)		50,000	50,000
mar-10	cash a/c --- Dr To Sales a/c (being good sold for cash)		1,00,000	1,00,000

mar-20	Salaries a/c. - - - Dr to cash a/c (being salaries paid for cash)	50,000	50,000
mar-31	Rent a/c - - - Dr to cash a/c (being rent paid for the month)	20,000	20,000
	Total	9,20,000	9,20,000

2. Journalise the following transactions in the book of ABC limited for the month of January 2023.

- ABC limited commenced into business with cash of rupees 8 lakh.
- 3rd january purchased machinery for cash 2 lakh.
- 5th january transport paid in cash 20,000/-
- 7th january sold goods to Shamp 10,000/-
- 8th january cash sales 50,000/-
- 12th january interest paid to Krishna 10,000/-
- 15th january dividend received 5,000/-
- dis 20th january, discount allowed 1,000/-
- 22nd jan, interest received 10,000/-
- rent paid to landlord (payment made Through cheque)

Date	particulars	lf	Dr	Cr
1st Jan 2023	Cash a/c ... Dr to ABC capital a/c (being business commenced with cash)		8,00,000	8,00,000
3rd Jan	Machinery a/c ... Dr to cash a/c (being machinery purchased for cash)		2,00,000	2,00,000
5th Jan	Transport a/c ... Dr to cash a/c (being trans- port paid incash)		20,000	20,000
4th Jan	Shyam a/c ... Dr to Sales a/c (being goods sold out to Shyam)		10,000	10,000
8th Jan	Cash a/c ... Dr to Sales a/c (being cash sales made)		50,000	50,000
12th Jan	Interest a/c ... Dr to cash a/c (being interest paid to Krishna)		10,000	10,000

15 th Jan	Cash a/c ... Dr to dividend (being dividend received)	5000	5000
20 th Jan	discount a/c ... Dr to cash a/c. (being discount allowed)	1000	1000
22 nd Jan	Cash a/c ... Dr to interest a/c (being interest received)	10,000	10,000
24 th Jan	Rent a/c ... Dr to Bank a/c (being rent paid to landlord payment paid through cheque)	15,000	15,000
	Total	11,21,000	11,21,000

3) Generalise the following transactions in the books of Ram for the month of march 2020.

- 1st march - Ram commenced business: Cash 5 lakhs machinery 3 lakhs, furniture 2 lakhs.
- 3rd march - cash deposited in bank 3 lakhs.
- 5th march - purchase goods from shyam 50,000
- 6th march - Goods return to shyam 5000
- 10th march - cash withdrawn from bank for personal use 10,000/-
- 15th march - Sold goods to krishna 30,000/-
- 18th march - Goods returned by krishna 5000/-

- 20th march - cash with drawn from bank for office use 20,000/-
- 20th march - Cash withdrawn from office for personal use 10,000/-
- 30th march - wages paid 20,000/-

Date	Particulars	Dr	Cr
1 st march 2020	cash a/c --- Dr 5,00,000 machinery a/c Dr 3,00,000 furniture a/c Dr 2,00,000 to Ram Capital a/c. (being Ram commenced with cash, machinery, furniture)	5,00,000 3,00,000 2,00,000	1,00,000
3 rd march	Bank a/c --- Dr to cash a/c (being cash deposited in bank)	3,00,000	3,00,000
5 th march	purchase a/c --- Dr to Shyam a/c (being goods purchased from Shyam on credit)	50,000	50,000
6 th march	Shyam a/c --- Dr to purchase return a/c (being goods return to Shyam)	5000	5000
10 th march	Drawings a/c --- Dr to bank a/c (being cash with drawn from bank for personal use)	10,000	10,000

15th march	krishna a/c --- Dr to Sales a/c (being goods sold to krishna on credit)	30,000	30,000
15th march	Sales return a/c --- Dr to krishna a/c (being goods return by krishna)	5000	5000
20th march	cash a/c Dr to bank a/c (being cash withdrawn from bank for office use)	10,000	10,000
30th march	wages a/c --- Dr to cash a/c (being wages paid)	20,000	20,000
	Total	1,45,000	1,45,000

4) Generalise the following transactions in the books for Suraj for the month of January 2023. 1st Jan - Suraj commenced business with cash 10 Lakhs.

3rd Jan - purchased machinery from Adithya payment paid through check.

5th Jan - Sold goods for cash 50,000

8th Jan - electricity paid through check 10,000

10th Jan - Insurance paid 5000.

12th Jan - Interest received from ABC limited 15,000.

14th Jan - rent received 10,000

16th Jan - Sold goods to Adhitya 10,000

18th Jan - Goods returned by Adhitya 2,000

20th Jan - Purchased furniture 11 Lakh

22nd Jan - Purchased land from Sundarain payment - paid through
Cheque 10,00,000

30th Jan - Dividend received 10,000

31st Jan - Discount received 5000

Date	Particulars	Dr	Cr
1 st Jan	Cash a/c ... Dr to Srijai capital a/c (being business commenced with cash)	10,00,000	10,00,000
3 rd Jan	Machinery a/c ... Dr to bank a/c (being purchase machinery from adhitya Payment paid through cheque)	2,00,000	2,00,000
5 th Jan	Cash a/c ... Dr to Sales a/c (being sold goods for cash)	50,000	50,000
5 th Jan	electricity a/c ... Dr to bank a/c (being electricity paid through check)	10,000	10,000
10 th Jan	insurance a/c ... Dr to cash a/c (being insurance paid)	5000	5000

12 th Jan	Cash a/c ... dr to interest a/c (being interest received from ABC limited)	15,000	15,000
14 th Jan	Cash a/c ... dr to rent a/c (being rent received)	10,000	10,000
16 th Jan	Adithya a/c ... Dr to Sales a/c (being goods sold to Adithya)	10,000	10,000
18 th Jan	Sales return a/c ... dr to Adithya a/c (being goods returned to adithya)	2000	2000
20 th Jan	Furniture a/c ... dr to cash a/c (being furniture purchased)	1,00,000	1,00,000
22 nd Jan	Land a/c ... dr to bank a/c (being land purchased from Subban payment through Cheque)	10,00,000	10,00,000
30 th Jan	Cash a/c ... Dr to dividend a/c (being dividend received)	10,000	10,000
31 st Jan	Cash a/c ... dr to discount a/c (being rent received)	5000	5000
	Total	24,00,000	24,00,000

5) Journalize the following transactions in the books of for the ABC limited month of march 2022:

- 1st march commenced business with cash 4 lakhs
 3rd march - purchase goods from Krishna 50,000/-
 5th march - purchased motor car for cash 2 lakhs
 10th march - purchased furniture for cash 20,000/-
 15th march - Goods withdrawn from office, for personal use 20,000 -
 20th march - Salaries paid through cheque 20,000
 25th march - advertisement paid through cheque 10,000
 30th march - cash sales 20,000
 31st march - Rent paid 10,000, wages paid 20,000, Salaries paid 10,000

Date	Particulars	Dr	Cr
1 st march 2023	Cash a/c... Dr to ABC limited Capital a/c (being business commenced with cash)	4,00,000	4,00,000
3 rd march	goods a/c... Dr to cash a/c (being goods purchased from Krishna)	50,000	50,000
5 th march	Motor a/c... Dr to cash a/c (being motor car purchased for cash)	2,00,000	2,00,000
10 th march	furniture a/c... Dr to cash a/c (being furniture purchased)	20,000	20,000

15 th march	Drawing a/c ... Dr to goods a/c (being goods withdrawn from office for personal use)	20,000	20,000
20 th march	Salary a/c ... Dr to bank a/c (being salaries paid through cheque)	10,000	10,000
25 th march	Advertisement a/c ... Dr to bank a/c (being advertisement paid through cheque)	10,000	10,000
30 th march	cash a/c ... Dr to Sales a/c (being cash sales)	20,000	20,000
31 st march	Rent a/c ... Dr wages a/c Sales a/c or to Cash a/c	10,000 20,000 10,000	40,000
	Total	10,40,000	10,40,000

prepare a paid balance from the following info as on 23 dec 2023

31st march 2023 - Capital 1,50,000/- Purchases 40,000 Cash in hand 7,000, cash at bank 8,500, electricity expenses 4800, Stationery 500, office equipment 2400, Sundry debtors 8000, machinery 1,60,000/- Salaries 11,400, Sales 1,05,400, Sundry creditor 5,000/-, rent 2,000, furniture 68,000, bank loan 10,000/-, investment 6,000, Commission paid 1,050, opening Stock 3,200, commission received 1,200, Postage and telegram 450.

Date	Particulars	Dr	Cr.
March 31	Capital		1,50,000
	Purchase	40,000	
	Cash in hand	7,000	
	Cash at bank	8,500	
	Electricity expenses	4,800	
	Stationery	500	
	Office equipment	2,400	
	Sundry debtors	8,000	
	Machinery	1,60,000	
	Salaries	11,400	
	Sales		1,05,400

Sundry creditors		5000
rent	2000	
furniture	16000	
Bank loan		10,000
investment	6000	
commission paid	1000	
opening stock	3200	
commission received		4200
Postage and telegram	450	
Total	2,71,600	2,71,600

• prepare a paid

Capital 45,000 (cr)	plant 75,000 d	General expenses 2,800 d
Sales 4,20,450 (cr)	trade expense 10,000 d	Advertiser 15,000 d
Creditores 15,000 (cr)	carriage inwards 2500 d	bills receivable 6,000 d
Interest rec 200 (cr)	carriage outwards 1,500 d	drawings 6,000 d
Bills payable 2000 (cr)	for office rent 1,500 d	Salaries 15,000 d
Cash at bank 12,500 (dr)	discount 350 d	wages 20,000 d
Cash in hand 2,000 (dr)	insurance 400 d	furniture 4,500 d
coal & gas 1000 (dr)	debtors 60,000 d	
Opening stock 45,000 (dr)	for off rent 3,000 d	
Purchase 2,25,000 (dr)	Printing 600 d	

10,23,500

20/12/2022

Ledgery

Ledger is the classification of account like personal, real and nominal accounts. It is mandatory for all the firms to maintain a set of accounts of the business according to the transactions of business.

Rules for ledger posting

- 1) If an account is debited in journal also be debited in ledger
- 2) If an account is credited in journal also be credited in ledger

Balancing the ledger account

It is mandatory to balance the ledger account at the end of the month by putting the excess amount either on the debit or credit side to close the ledger accounts.

It is customary to use the words "to" and "by" while posting the general transactions into respective ledger accounts.

Trial balance

Trial balance is prepared for accuracy purpose every month end by extracting the balances of ledger accounts.

Rules of Trial balance

Debit expenses losses and assets.

Credit income gains and liability

Examples of

Cash in hand, cash at bank, marketable securities, patents, machinery, equipment, furniture, land and buildings, goodwill, bill receivable, prepaid expenses, etc.

Liabilities

capital, outstanding expenses, bills payable, credit, bankover, draft, bank loan etc,

Journalize the following transactions post them into ledger accounts and prepare trial balance.

1st march 2020 - Cash Sales 20,000

5th march - Salaries paid 20,000

20th march - Cash Sales 30,000

Journal

Date	Particulars	Jf.	Dr	Cr
mar 01	cash alc Dr to Sales alc (being cash sales made)		20,000	20,000
mar 05	Salaries alc ... Dr to cash alc (being salaries paid in cash)		20,000	20,000
mar 20	cash alc ... Dr to Sales (being cash sales made)		30,000	30,000
	Total		70,000	70,000

Dr.

Cash account

Cr.

Date	Particulars	Jf	Rs	Date	Particular	Jf	RS
mar 01	To Sales		20,000	mar 05	By Salaries		20,000
mar 20	To Sales		30,000	mar 31	By balance b/d		30,000
			<u>50,000</u>				<u>50,000</u>
April - 01	To balance credit		30,000				

Dr.

Sales account

Cr.

Date	Particulars	Jf	RS	Date	Particular	Jf	RS
mar - 31	to balance b/d		<u>50,000</u>	mar - 01	By cash		20,000
			50,000	mar - 20	By cash		30,000
							<u>50,000</u>
				April - 01	By balance c/d		50,000

Salaries account

Date	Particulars	Jf	Rs	Date	Particulars	Jf	Rs.
March - 05	To cash		<u>20,000</u>	mar - 31	By balance		<u>20,000</u>
			20,000				20,000
April - 01	To balance c/d		20,000				

Trial balance

Date	Particulars	Dr	Cr
March - 31	cash	30,000	
March - 31	sales		50,000
March - 31	Salaries	20,000	
	Total	50,000	50,000



Following balances as on 31st March 2020 prepare trial balance

Interest on Capital - 24,000/-	Capital - 8,00,000/-
Creditors - 6,00,000/-	Drawings - 2,46,000/-
Discount received - 23,000/-	Machinery - 3,00,000/-
Loan - 1,44,000/-	Bills payable - 30,000/-
Purchase return - 40,000/-	Furniture - 6,00,000/-
Sales return - 6,000/-	Debtors - 5,00,000/-
Advertisement - 1,63,000/-	Bank loan - 2,00,000/-
Commission received - 20,000/-	Patents - 60,000/-
Rent - 10,000/-	Opening Stock - 12,00,000
Purchase - 19,00,000/-	Sales - 32,60,000/-
Salaries - 1,23,000/-	

Trial balance on 31-03-2020

Date	Particulars	Dr.	Cr.
Mar-31	Interest on Capital	24000	
	Creditors		6 00 000
	Discount received		23,000
	Loan		1 44,000
	Purchase return		40,000
	Sales return	6000	
	Advertisement	1 63,000	
	Commission received		20,000
	Rent	10,000	
	Purchase	19,00,000	
	Salaries	1,23,000	



Capital		8,00,000
Drawings	2,46,000	
Machinery	3,00,000	
Bills payable		20,000
Furniture	6,00,000	
Debtors	5,00,000	
Bank loan		2,00,000
Patents	60,000	
Opening stock	12,00,000	
Sales		32,60,000
Total	51,34,000	51,34,000

Prepare trial balance 31-Jan-2023

Furniture - 15,000/- (dr)

Capital - 54,000 (cr)

Cash in hand - 3,000 (dr)

Opening stock - 50,000 (dr)

Fixed deposits - 1,34,000 (dr)

Drawings - 5,000 (dr)

Provision for bad debts - 3,000/- (cr)

Cash at bank - 10,000 (dr)

Purchase - 3,00,000/- (dr)

Salaries - 19,000/- (dr)

Carriage inwards - 41,000/- (dr)

Insurance - 6,000/- (dr)

Rent - 22,000/- (dr)

Sundry debtors - 40,000/- (dr)

Sales - 6,00,000/- (cr)

Advertisement - 10,000/- (dr)

Postage and telegram - 3,400 (dr)

Bad debts - 2,000/- (dr)

Printing and Stationary - 9,000/- (dr)

General expenses - 13,000/- (dr)

Sundry Creditors - 40,000/- (cr)

Deposit from customer - 6,000/- (cr)

Unit 4

1) From the following info received from the books of MR. VENU prepare trading, profit & loss acc in the year end 31 dec 2020 and balance sheet as on date.

Debit

Capital	45,000/-	
Interest received	200/-	
(Bills payable)		
Cash at bank	12,500	(BA)
Cash at hand	2000	(BA)
Coal and gas	1000	(TA)
Purchasing	10,000	
Plant	22,500	(BA)
opening stock	15,000	(TA)
purchases	2,25,000	(BA) TA
plant	45,000	(BA)
Trade expenses	10,000	P&L
Carriage inwards	2,500	TA
Carriage outwards	1,500	P&L
factory Rent	1,500	TA

Credit

Sales	4,20,450/-	(TA)
creditors	15,000/-	(BL)
Bills payable	2,000/-	(BL)
Capital	45,000/-	(BL)
Interest received	200/-	(P&L)
Discount	350	(P&L)
Insurance	700	(P&L)
debtors	60,000	(BA)
officerment	300	P&L
printing	600	P&L
General expenses	2,800	P&L
Adv	15,000	P&L
Bills receivable	6,000	(BA)
D. drawings	600	(BL)
Salaries	10,000	P&L
Wages	2,000	TA
factory	7,000	(BA)

Adjustments:

closing Stock is valued at 35,000/-

Depreciation on plant at 10 %

Outstanding factory rent 300

" office rent 600

Sol:

Trading and P&L a/c for the year end 31-12-2023

<u>Particulars</u>	<u>Rs</u>
To opening Stock	45000
To purchases	225000
To coal & gas	1000
To carriage inwards	2500
To factory rent	1500
Add outstanding	300
	<u>1800</u>
To wages	201000
To Gross profit	458750 -
	<u>160450</u>

<u>Particulars</u>	<u>Rs</u>
By Sales	420750
By closing Stock	35000
	<u>455750</u>



To Trade expens 10,000
 To cartage outwards 1500
 To discount 350
 To insurance 400
 To office rent 3000 3600
 Add outstanding 600

To printing 600

To General expenses 2800

To Advertisement 15000

To salaries 15000

To depreciation on 45000

plant @ 10% 75000 x 10%

To Net profit b/d 103600

Carried over to Capital A/c

160650

By Gross profit old 160450

By interest receivable 200

160650

Balance Sheet as on 31-12-2020

Liabilities	Rs	Assets	Rs
Capital	45000	Cash at bank	12500
Add. Net profit	103600	Cash in hand	2000
	<u>148600</u>		



less: Drawings 6000	142600	less: Dep @ 10% 7800	67500
Creditors	15000	Debtors	60000
Bills payable	2000	Bills receivable	6000
outstanding factory rent	300	furniture	78000
outstanding office rent	600	closing stock	38000
Total	190500		190500

Assignment

Q From the following information, prepare a trading, P&L account for the year ended 31st dec 2006 & balance sheet as on table. (03 Jan 2024)

Capital - 30000 (B)	OP with central bank - 6,000 (B)
Drawing - 6000 (B)	Purchases - 50,000 (T)
Sundry creditors - 43000 (B)	Duty & cleaning - 3,500 (T)
Bills payable - 4000 (B)	Sales - 1,28,000 (T)
Sundry debtors - 51,000 (B)	Salaries - 9,500 (P&L)
Bills receivable - 5000 (B)	Sale's return - 1000 (T)
loan & advance to deepak & home fitting - 12000 (B)	Purchase return - 1100 (T)
Opening stock - 44,000 (T)	Commission/ travel expenses - 4400 (P&L)
Cash in hand - 900 (B)	General expenses - 2500 (P&L)
Cash at bank - 12,500 (B)	Rent paid - 2000 (P&L)
	Commission received - 4000 (P&L)

Adjustments

- closing stock is valid at 50,000
- Interest to be received 200,000
- outstanding Salaries 500,000
- Depreciation on Fitting 10%
- commission received in advance 600
- Allow interest on Capital 8%

Sol:

(1) 500,000 - 500,000 = 0

(2) 200,000 - 200,000 = 0

(3) 10% of 50,000 = 5,000

(4) 600 - 600 = 0

(5) 8% of 500,000 = 40,000

(6) 10% of 50,000 = 5,000

(7) 500,000 - 500,000 = 0

(8) 200,000 - 200,000 = 0

(9) 50,000 - 50,000 = 0